Lesson 4: Combustion and the Fire Triangle

4.1 What combustion is

Combustion is a chemical reaction in which a fuel reacts rapidly with oxygen and releases energy as heat and light. The reaction is exothermic, meaning it gives out more energy than it takes in. Complete combustion of a hydrocarbon produces carbon dioxide and water vapour. Incomplete combustion, where oxygen is limited, produces carbon monoxide and soot instead, which is why a yellow smoky flame indicates a poor air supply.

4.2 The fire triangle

Three things must be present for combustion to begin and continue: fuel, oxygen, and heat. These are drawn as the three sides of the fire triangle. Remove any one side and the fire goes out. Water cools a fire, removing heat. A fire blanket smothers a fire, removing oxygen. Turning off a gas tap removes the fuel. This is the principle behind every method of extinguishing a fire.

4.3 Activation energy

A fuel and oxygen sitting together do not ignite on their own at room temperature. Energy must first be supplied to start the reaction; this minimum energy is called the activation energy. A spark or a lit match provides it. Once burning has started, the heat released supplies the activation energy for the next portion of fuel, so the reaction sustains itself until the fuel or the oxygen runs out.